

Make sure to show all work to receive full credit

Score: _____ out of 48

Important: Leave all answers as an **exact** value unless otherwise stated.

1. (3 points) Solve $\log_4(x - 2) = 5$ for x .

2. (5 points) Solve $2 \cdot 4^{3x-4} = 10$.

3. (3 points) On the day of a child's birth, a parent deposits \$10,000 in a trust fund that pays 10% interest, compounded continuously. Determine the balance in this account on the child's 18th birthday. (Round your answer to two decimal places.)

4. (5 points) Determine if $\left(\frac{\sqrt{13}}{7}, \frac{6}{7}\right)$ is on the unit circle.

5. (5 points) Determine the principal P that must be invested at rate $r = 8\%$, compounded monthly, so that \$500,000 will be available for retirement in $t = 40$ years. (Round your answer to the nearest cent.)

6. Perform the conversion below.

a. (2 points) Convert 135° degrees to radians

b. (2 points) Convert $\frac{2\pi}{3}$ radians to degrees

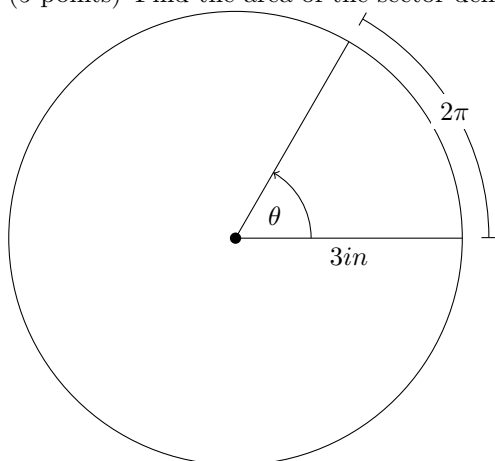
7. (2 points) Find any angle (other than the angle itself) that is coterminal with $\frac{7\pi}{6}$.

8. (3 points) A $7\frac{1}{4}$ -inch circular power saw blade rotates at 5200 revolutions per minute. Find the linear speed (in feet per minute) of the saw teeth as they contact the wood being cut.

9. (3 points) Evaluate $\cos\left(\frac{7\pi}{6}\right)$.

10. (5 points) Write $3\ln(x) + \frac{1}{2}\ln(y) - 4\ln(x^2 + y)$ as a single logarithm.

11. (5 points) Find the area of the sector defined by θ below.



$$\frac{1}{x} \Big|_{x=1}$$

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12. (5 points) Find all vertical, horizontal and/or oblique (slant) asymptotes that exist of the graph of the function $f(x) = \frac{x^2 - 7x - 8}{4x^2 + x - 3}$.

Formulas

$$A = P \left(1 + \frac{r}{m} \right)^{mt}$$

$$A = Pe^{rt}$$

$$A = \frac{1}{2} r^2 \theta$$

$$\omega = \frac{\theta}{t}$$

$$v = \frac{\theta \cdot r}{t}$$

$$n(t) = n_0 e^{rt}$$

$$m(t) = m_0 e^{-rt} \text{ where } \frac{\ln(2)}{h}$$

$$\log_b z = y \text{ iff } b^y = z$$

